



ADAMA SCIENCE AND TECHNOLOGY UNIVERSITY

School of Electrical Engineering and Computing

Department of Computer Science and Engineering

Course: programming languages

Title: Report on Tebeblang

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TebebLang

Design Decisions

TebebLang was designed using Amharic keywords to help Ethiopian beginners learn programming concepts more naturally. Python was selected as the target language because it is simple, readable, easy to execute, and provides strong Unicode support.

The implementation uses a real programming language architecture consisting of a lexer, parser, AST, and code generator instead of simple text replacement.

Why the Language Helps Beginners

The language helps beginners by using familiar local-language keywords and simplified syntax. The structure avoids excessive punctuation and resembles natural reading patterns, making programs easier to understand.

Features that improve beginner accessibility include:

- Amharic keywords
- readable syntax
- reduced punctuation
- Simple control flow structures

Challenges

The main challenges during implementation included:

- handling Unicode Amharic characters correctly
- generating valid Python indentation
- designing readable local-language keywords
- building parser and AST structures without ambiguity